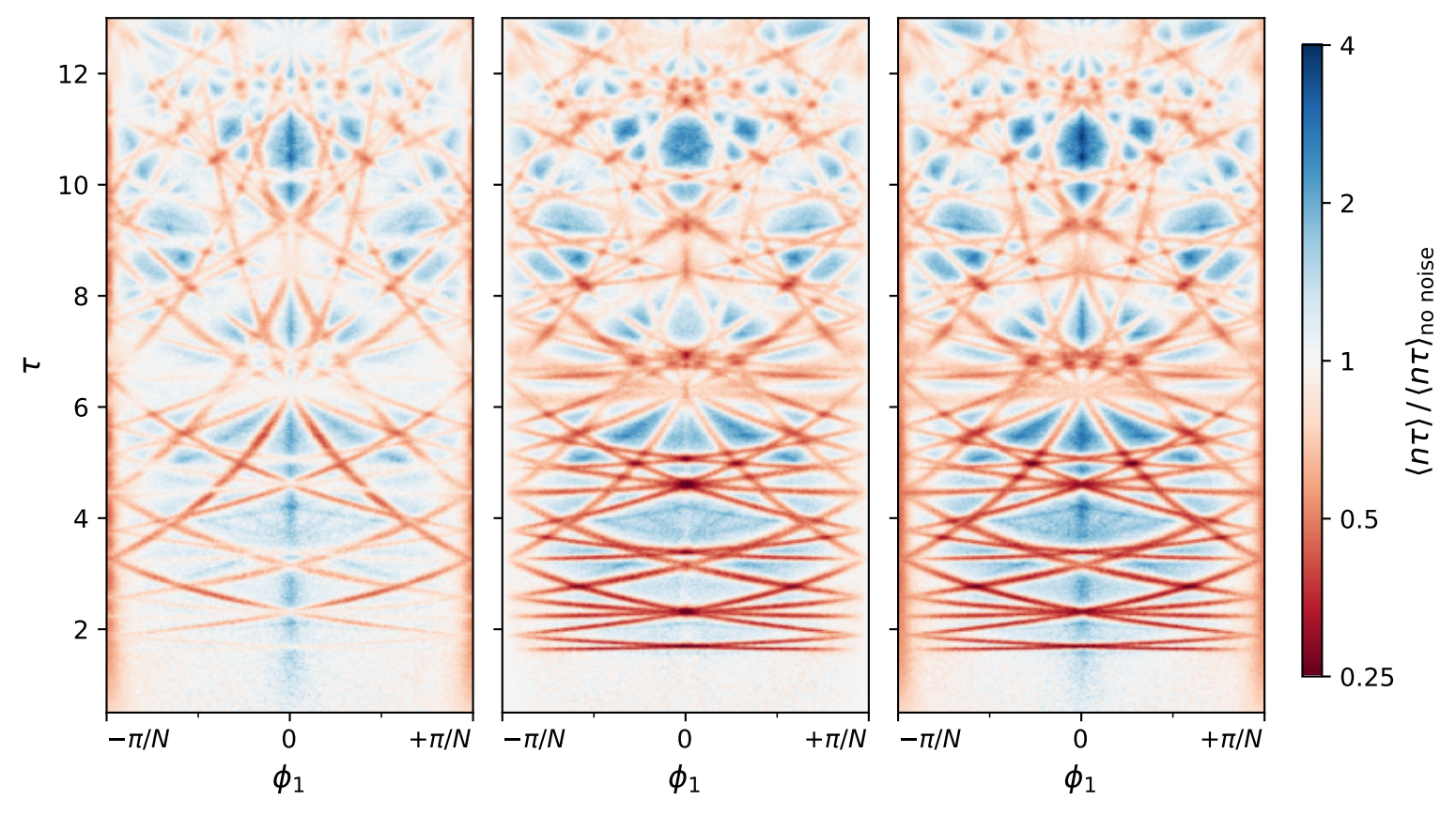

τ
12
10
8
6
4
2
−π/N
0
+π/N
ϕ1
4
2
1
0.5
0.25
⟨nτ⟩ / ⟨nτ⟩no noise